CMP682 — Final Project

MissVARPath

Missense Variant Pathogenicity Prediction

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Three questions

Every missense variant goes through a battery of in-silico predictors — PolyPhen-2, SIFT, REVEL, CADD, AlphaMissense, DITTO. Each disagrees with the others; most were trained on overlapping ClinVar data — the same database we use to label our targets.

1. Can a meta-classifier do better than any single predictor by combining them?

2. How much of any improvement is data-circularity — predictors trained on ClinVar, evaluated on ClinVar?

3. Can we predict pathogenicity from raw data alone — without using any pre-existing predictor's score?

Data

21,872

ClinVar / OpenCRAVAT missense variants

5,468 × 4

perfectly balanced 4-class — Benign / Likely-benign / Likely-pathogenic / Pathogenic

777 → 208

raw schema → numerical features after preprocessing

5,428

strict-VUS rows in the curated pro-set

Four task definitions used in the project

4-class

original ClinVar label space

2-class

Likely-* collapsed into definitive labels → 10,936 / 10,936

3-class (with VUS)

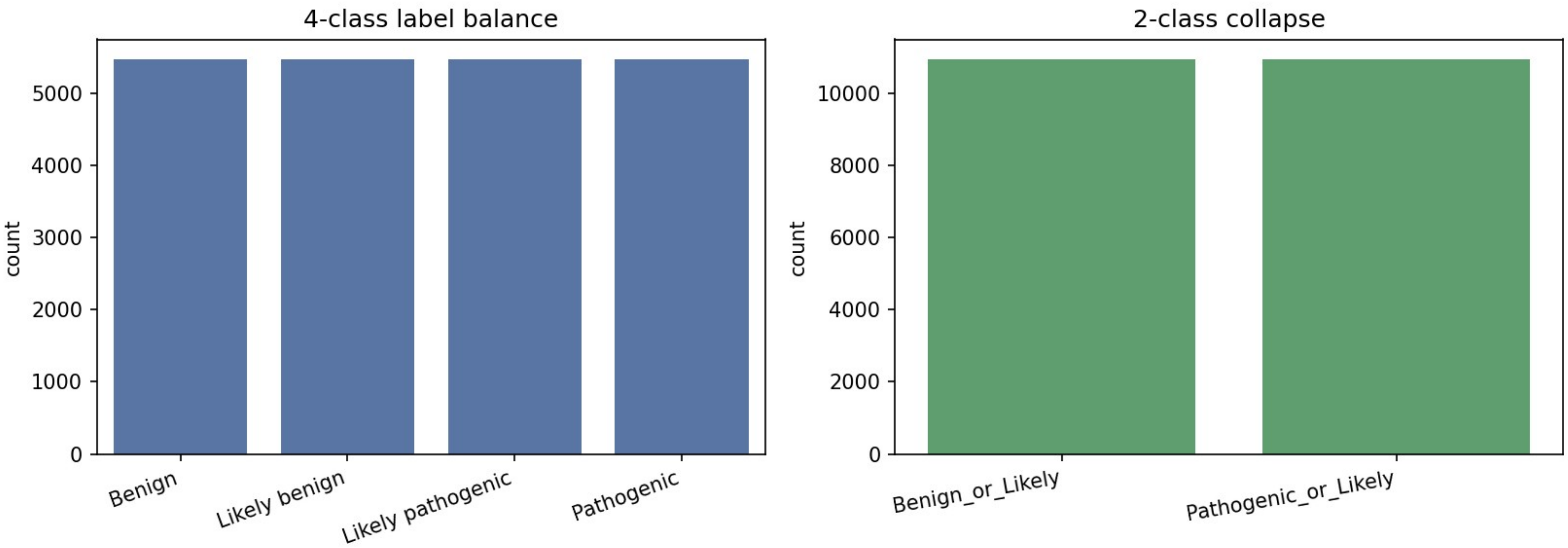
Benign-side / Pathogenic-side / VUS

5-class (with VUS)

all four ClinVar labels + VUS — essentially balanced

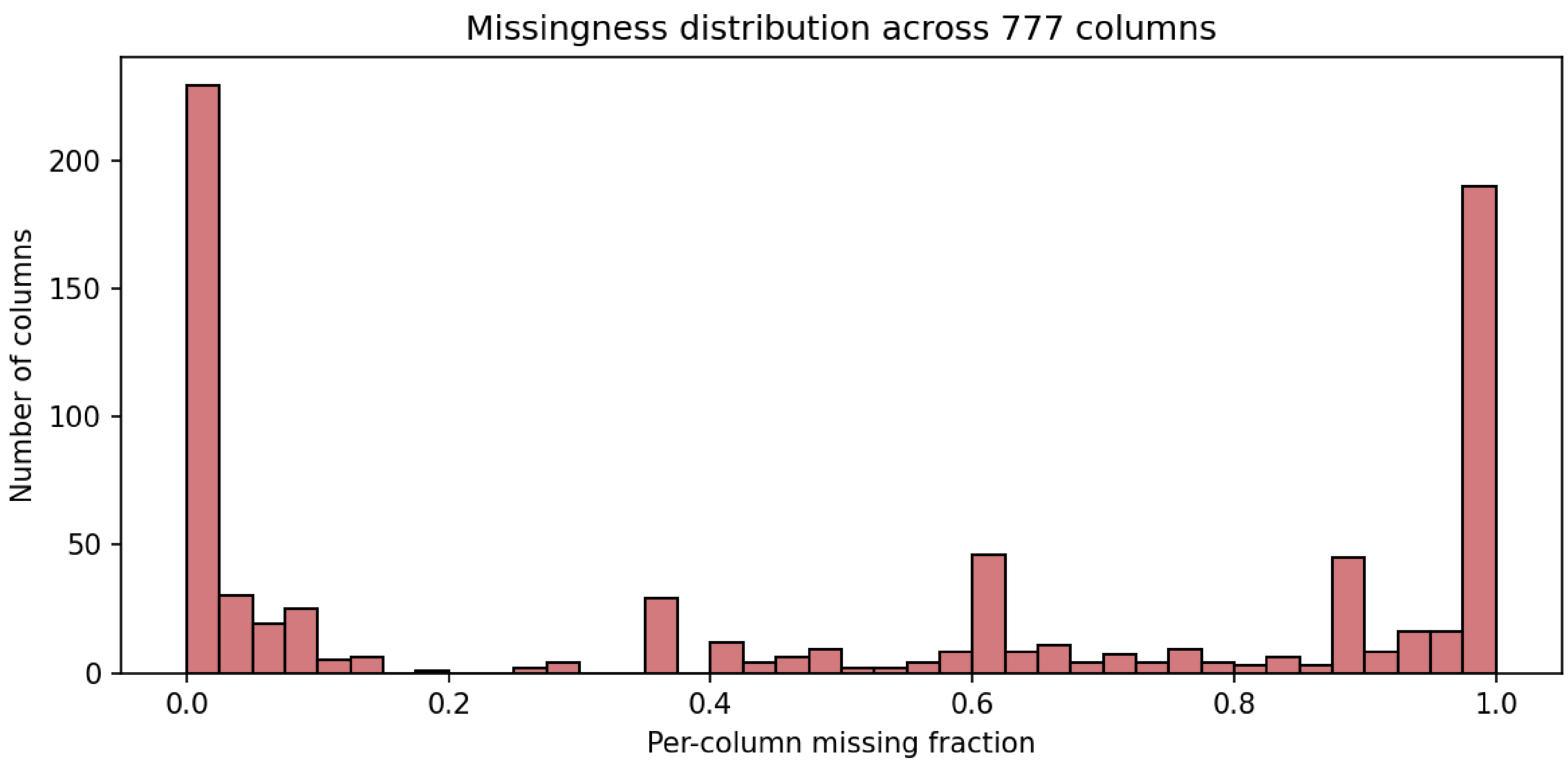
Label balance — by design

5,468 rows per ClinVar class in the 4-class space; 10,936 / 10,936 in the 2-class collapse. Any class-imbalance effect is gone by construction.



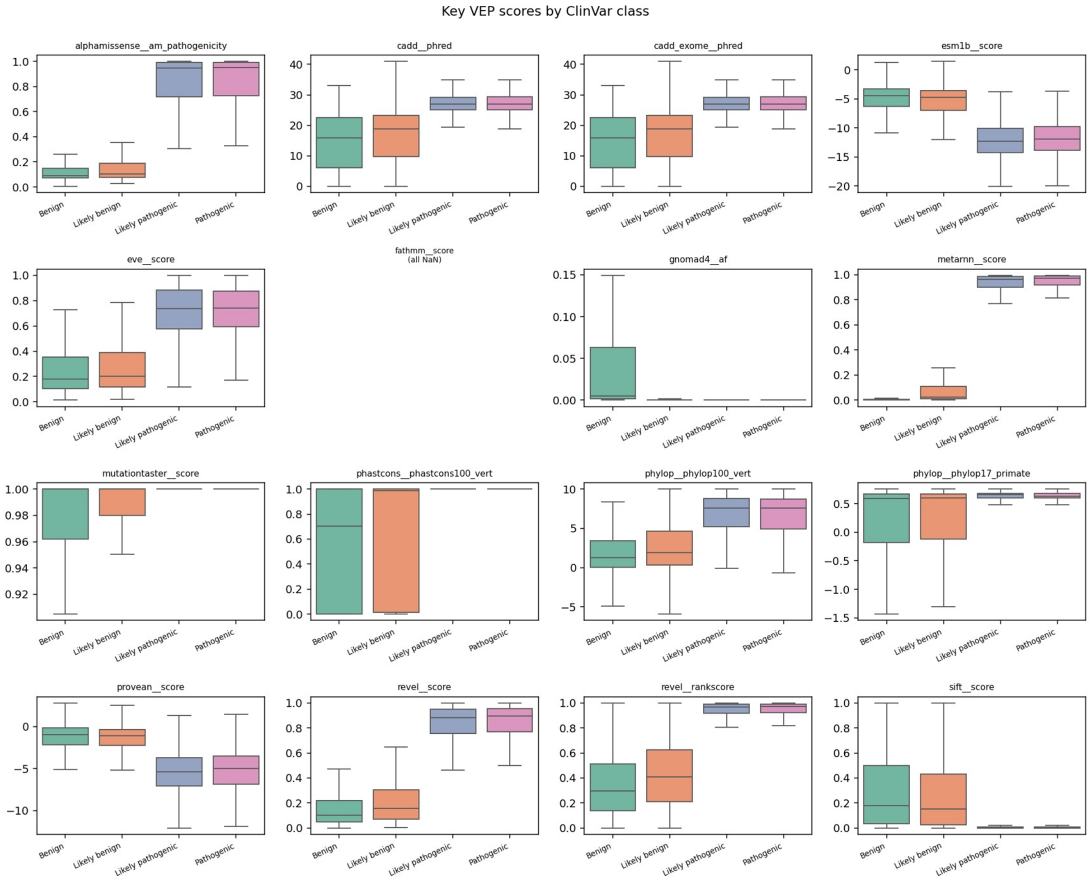
Missingness across the 777-column raw schema

Per-column missing fraction. Everything above the 50% threshold is dropped during preprocessing — 134 columns lost that way. 208 numerical features survive.



Do the predictor scores actually separate classes?

Boxplots of headline VEP scores stratified by the 4-class label. DITTO, MetaRNN, REVEL, AlphaMissense all separate cleanly; gnomAD AF runs the other way — high frequency means benign (the ACMG BS1 rule).



Methods

Model suite — 11 estimators

Classical / linear

KNN
NearestCentroid
CosineSimilarity (custom)
DecisionTree
LDA
QDA
LinearSVC
RidgeClassifier
SGDClassifier

Ensembles

AdaBoost
HistGradientBoosting

Evaluation regimes — 5 cuts

Canonical	stratified 80/20 holdout + 5-fold CV
Augmented	+ 196 k-mer + 10 BLAST locus-neighbour features
Gene-stratified	no gene appears in both train and test folds
No-VEP / VUS	drop every learned-predictor + conservation score
Raw-only	keep only population AF, position, sequence composition

RANDOM_STATE = 42 throughout. Per-model grid-search tuning by 5-fold CV macro-F1; AdaBoost grid stopped at combo 23/27 (compute budget).

Headline numbers

4-class macro-F1

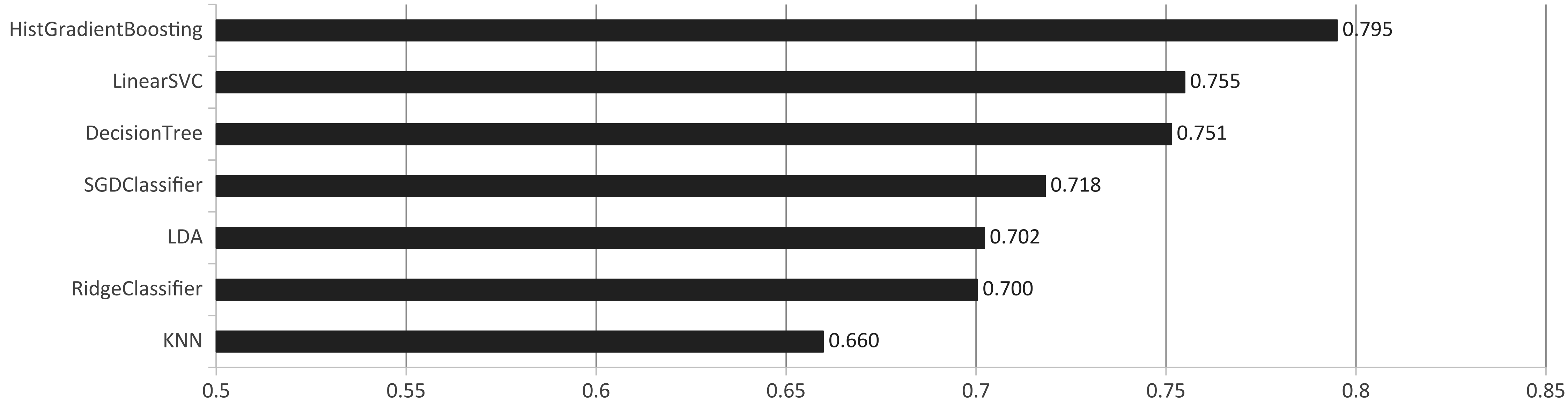
0.795

HistGradientBoosting · 21,872 variants · 208 features

2-class macro-F1

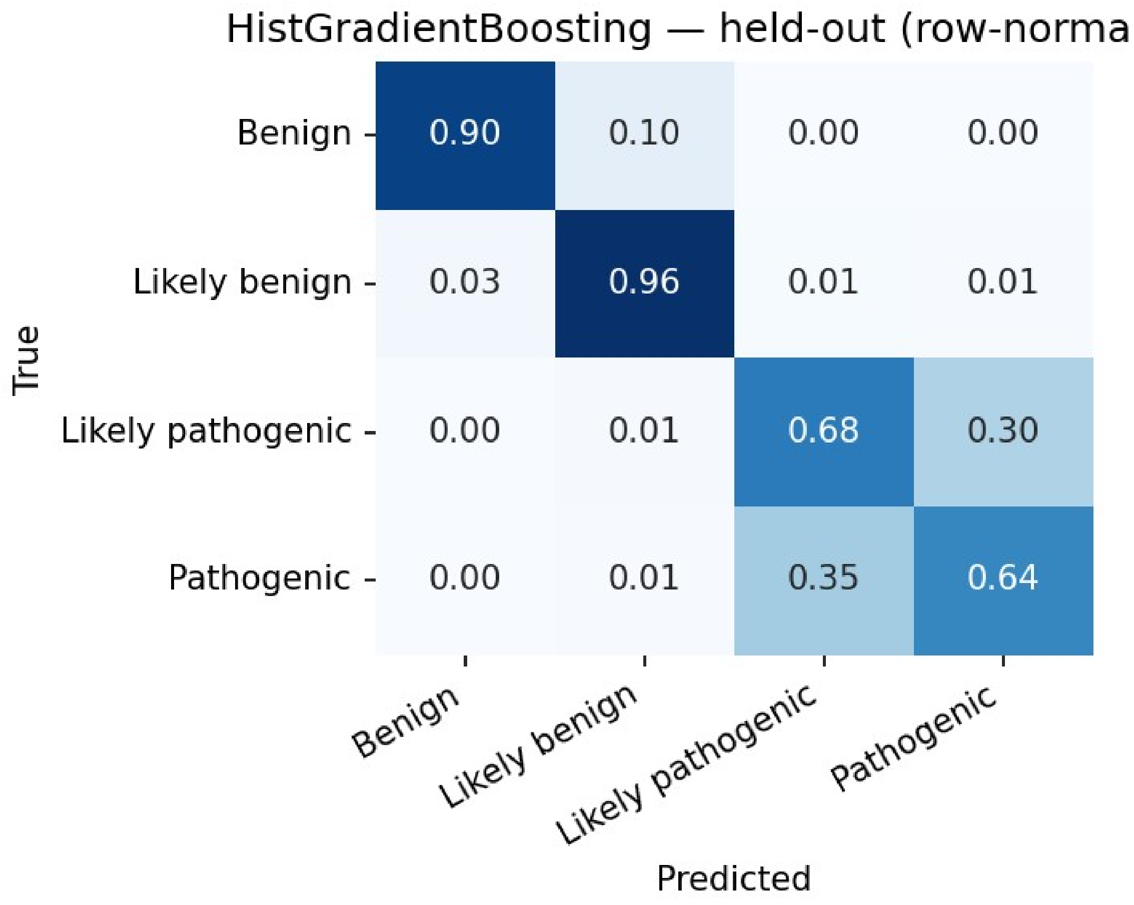
0.987

HistGradientBoosting · 10,936 / 10,936 collapse



Where the error lives — 4-class confusion matrix

Held-out, row-normalized. Definitive Benign / Pathogenic call near-perfect (>0.93 / 0.79). Residual difficulty is the Pathogenic ↔ Likely-pathogenic boundary.



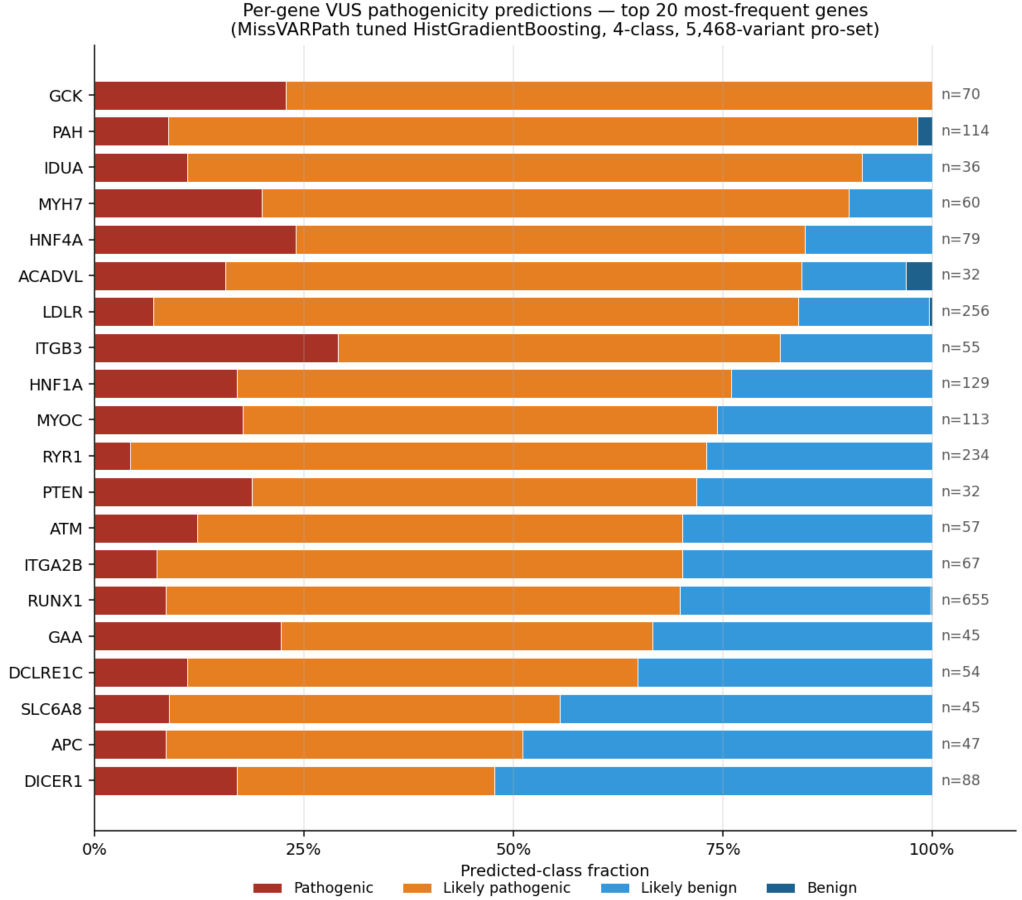
Stress tests — bounding the data-circularity

Regime	4-class F1	2-class F1
Canonical tuned (headline)	**0.795**	**0.987**
Augmented (k-mer + BLAST)	0.795	0.990
Gene-stratified CV	0.766	0.987
No-VEP, k-fold	0.773	0.976
No-VEP, gene-stratified	0.761	—
Raw-only (no predictor, no conservation)	0.721	0.934
DITTO removed	0.791	—

Graceful degradation across regimes: HistGB loses 0.027 to gene-stratified CV, 0.022 to no-VEP, 0.074 to raw-only. The 2-class task is essentially insensitive to all stresses.

Real VUS deployment — per-gene predictions

Top-20 most-frequent genes in the 5,468-variant pro-set, stacked by 4-class prediction. GCK 100%, PAH 99%, MYH7 90%, LDLR 84% pathogenic-side.



VUS deployment — the four numbers

Applying the tuned HistGradientBoosting to 5,468 strict-VUS variants annotated through the same pipeline.

62% / 38%

pathogenic-side / benign-side split on 2-class head

80%

of predictions have max probability > 0.9

87%

of 4-class predictions hedge into Likely-* — only 13% definitive

95.4%

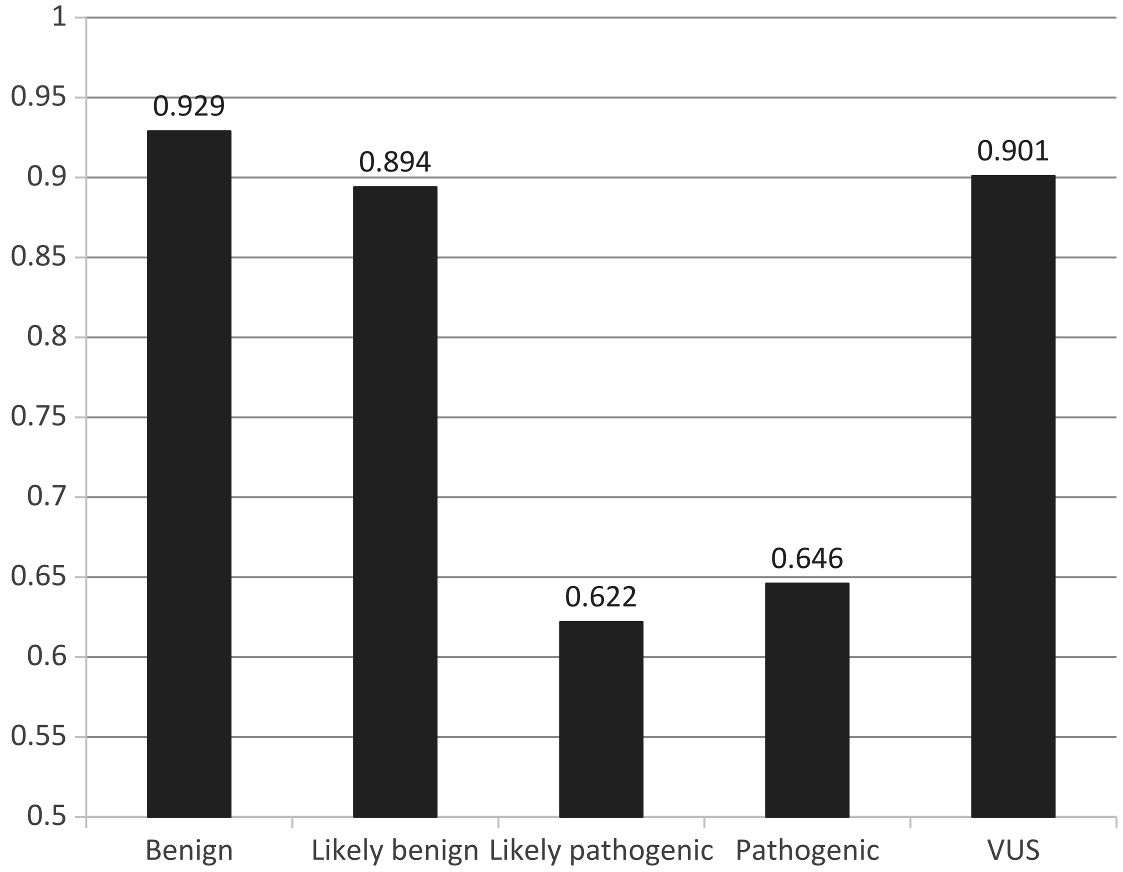
cross-task agreement (4-class collapsed vs. 2-class head)

VUS as a class

With VUS added as a labelled class

2-class no-VUS	**0.987**
3-class with VUS	**0.946**
4-class no-VUS	**0.795**
5-class with VUS	**0.798**

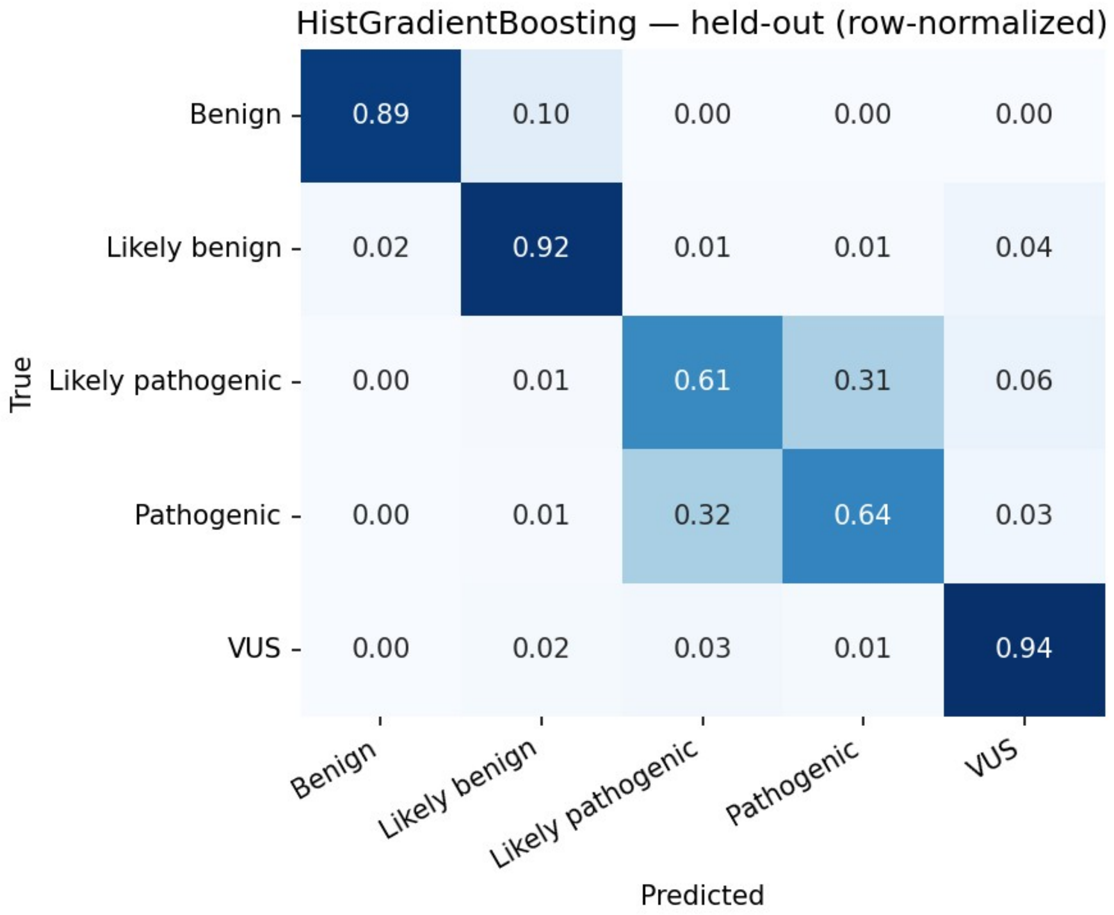
Adding VUS as a 5th class does NOT hurt overall macro-F1 — it slightly improves it. MCC improves by +0.022.



VUS reaches F1 = 0.901 — between Benign and Likely-benign in difficulty. One of the cleanest classes to recognise.

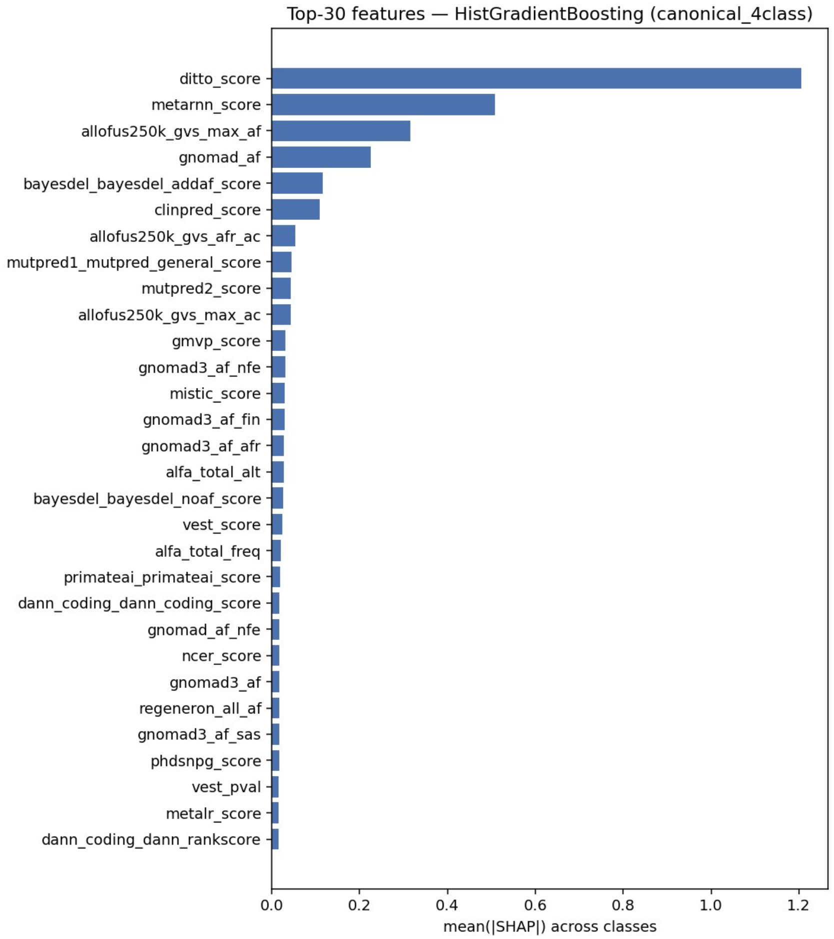
5-class confusion matrix — where VUS sits

Held-out, row-normalized. VUS row reaches 0.94 diagonal — cleaner than Pathogenic (0.64) or Likely-pathogenic (0.61). VUS misclassifications hedge into Likely-, not into definitive calls.*



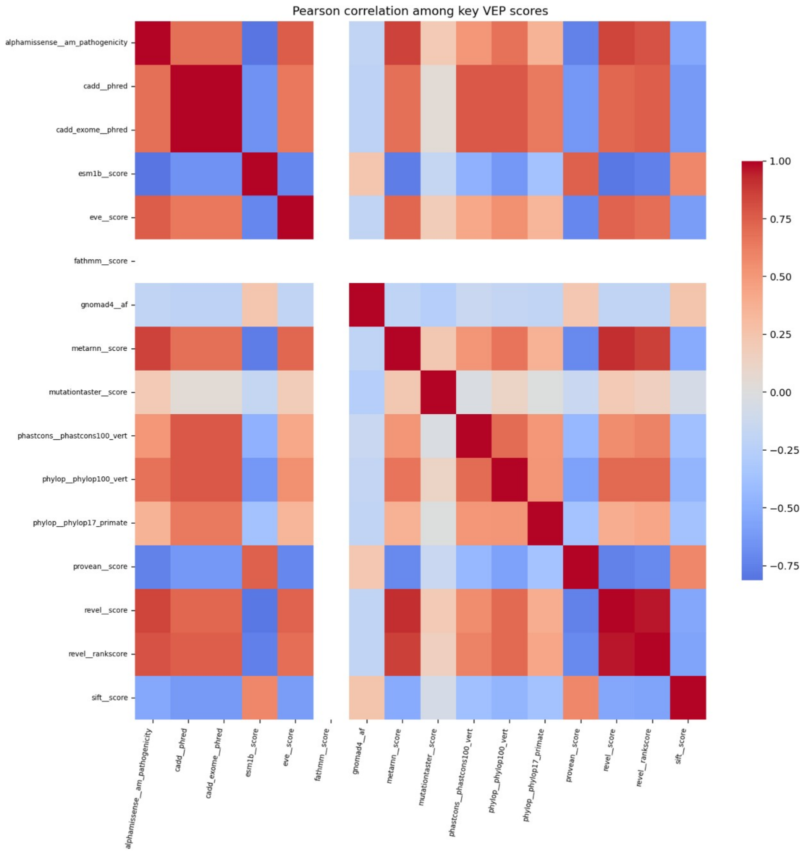
What drives the booster — SHAP top-30

Canonical 4-class HistGradientBoosting. DITTO leads, MetaRNN second, then population allele frequencies — the model is using both predictor scores and frequency data.



Predictor correlation — the redundancy structure

Pearson correlation among key VEP scores + gnomAD AF. Predictor scores form a tight cluster; gnomAD AF sits in its own corner — the structural reason predictors substitute for each other, but population AF does not.



Useful vs. waste — feature ablation

Group	4-class	2-class	5-class	3-class	n feat
population_af	+0.0471	+0.0025	+0.0405	+0.0054	47
functional	-0.0012	-0.0005	+0.0266	+0.0469	4
other_vep	+0.0133	-0.0002	+0.0137	+0.0073	56
cadd	-0.0054	-0.0005	-0.0001	-0.0015	4
ditto	+0.0039	+0.0046	+0.0001	+0.0033	1
metarnn	+0.0014	+0.0005	+0.0045	+0.0005	2
bayesdel	-0.0045	-0.0002	-0.0004	-0.0003	4
position	-0.0044	+0.0002	+0.0018	+0.0029	2
revel	-0.0018	+0.0005	-0.0044	-0.0015	2
chasmplus	+0.0026	+0.0011	+0.0028	-0.0002	68
alphamissense	-0.0015	+0.0005	-0.0028	-0.0017	1
conservation	+0.0018	-0.0007	-0.0001	+0.0000	17

Greyed rows: HistGB $|\Delta| < 0.005$ across every task. 8 of 12 groups (97 of 208 features) are waste at the group level.

Per-feature permutation — the redundancy story

Removing a 'waste' group is cheap — but individual features inside those groups are still highly predictive. Group LOO measures replaceability, not informativeness.

ditto_score

0.133

individual macro-F1 weight (4-class)

metarnn_score

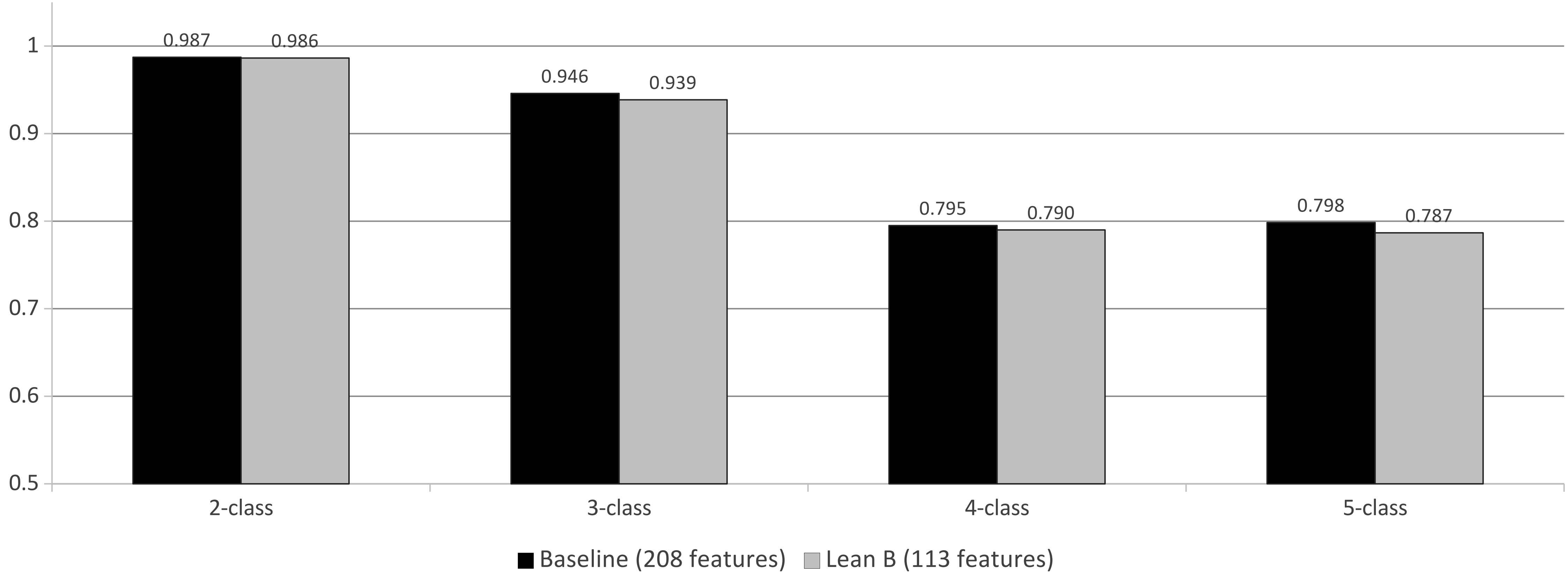
0.048

individual macro-F1 weight (4-class)

Inter-predictor redundancy. Remove DITTO and MetaRNN compensates; remove MetaRNN and DITTO compensates. Only by removing several correlated meta-classifiers together does the redundancy buffer collapse — which is what Lean B exploits.

Lean B — 45% smaller, 0.012 cost

Lean B drops 95 of 208 features, keeping ditto_score and metarnn_score (the per-feature winners).



At most 0.012 macro-F1 cost on any task. Several models (KNN, NearestCentroid, CosineSimilarity) actually improve on the lean subset.

Future directions

- **Cross-database validation**

  replicate Lean B numbers on HGMD or a held-out ClinVar snapshot

- **True homology BLAST**

  blastp on UniRef50 with the amino-acid substitution applied

- **Finish the AdaBoost tuning grid**

  the partial 23/27 grid was halted for compute budget

- **Probability calibration**

  Platt or isotonic regression so predict_proba is a usable risk score

- **Gene-stratified VUS-as-class**

  test whether VUS-detection survives without same-gene leakage

- **Variant-level error analysis**

  manual review of the Pathogenic ↔ Likely-pathogenic confusion region

Thank you

Questions?

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github.com/doruktopcu/MissVarPath-Missense-Detection